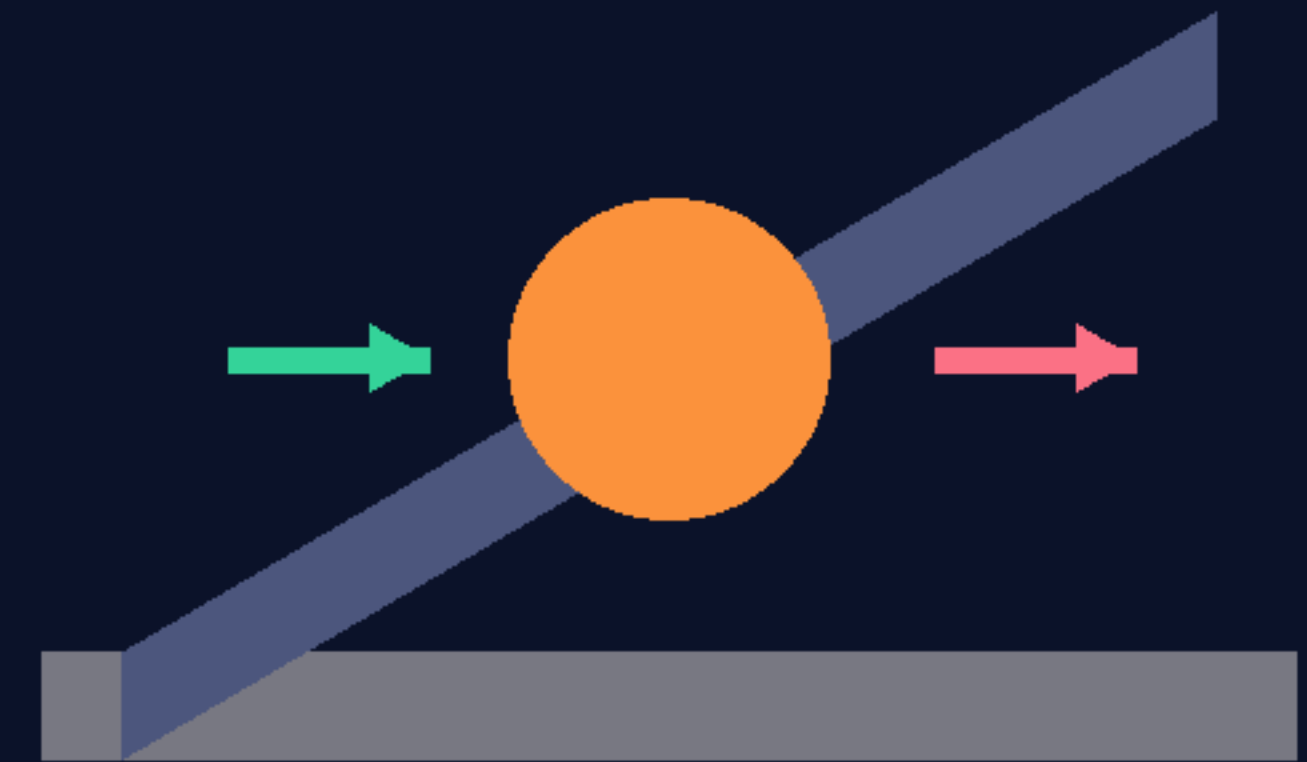


Forces Around Us

A force is a push or a pull. You use force when you push a door, pull a drawer, kick a ball, lift a bag, or press clay. Force is part of almost every movement around us.

PUSH, PULL, FRICTION

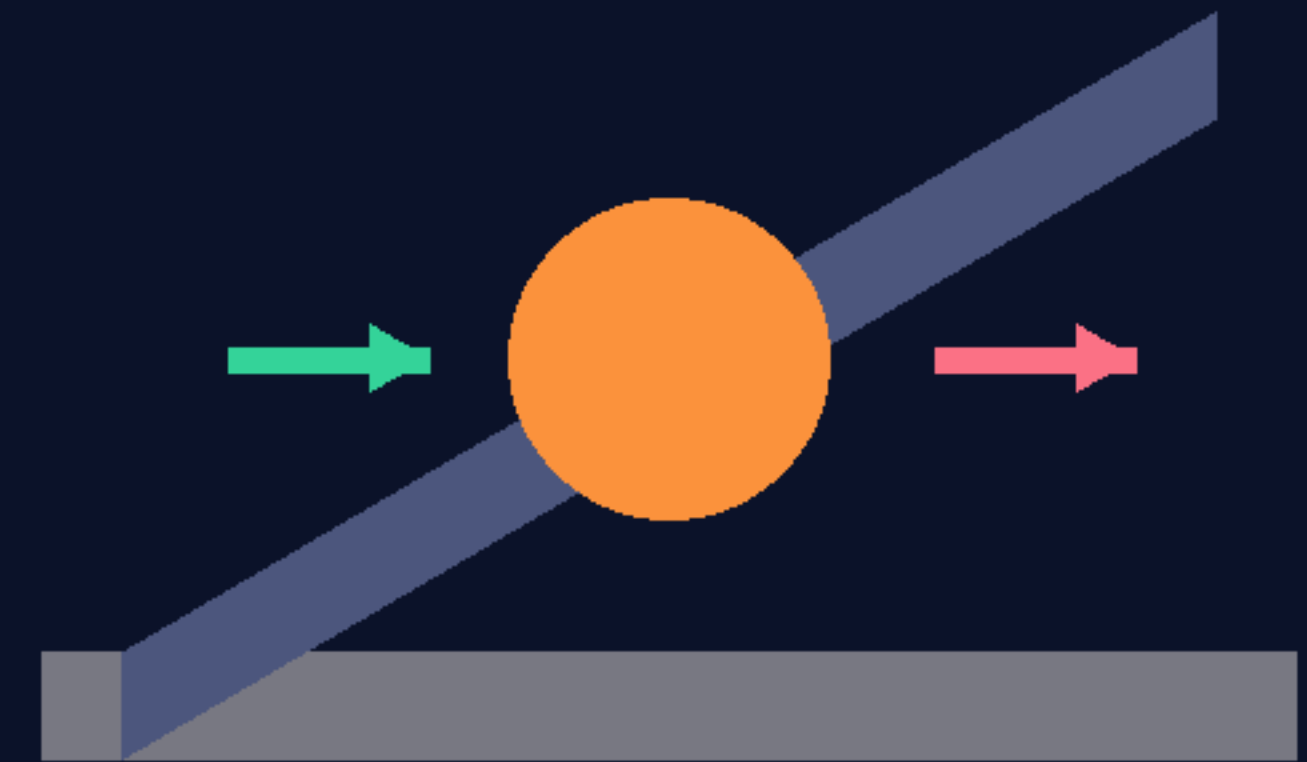


Quick check: Show one push and one pull action.

Push and Pull

A push moves something away from you. A pull moves something toward you. Opening a drawer is a pull. Closing a door can be a push. Try finding five push and pull actions at home.

PUSH, PULL, FRICTION

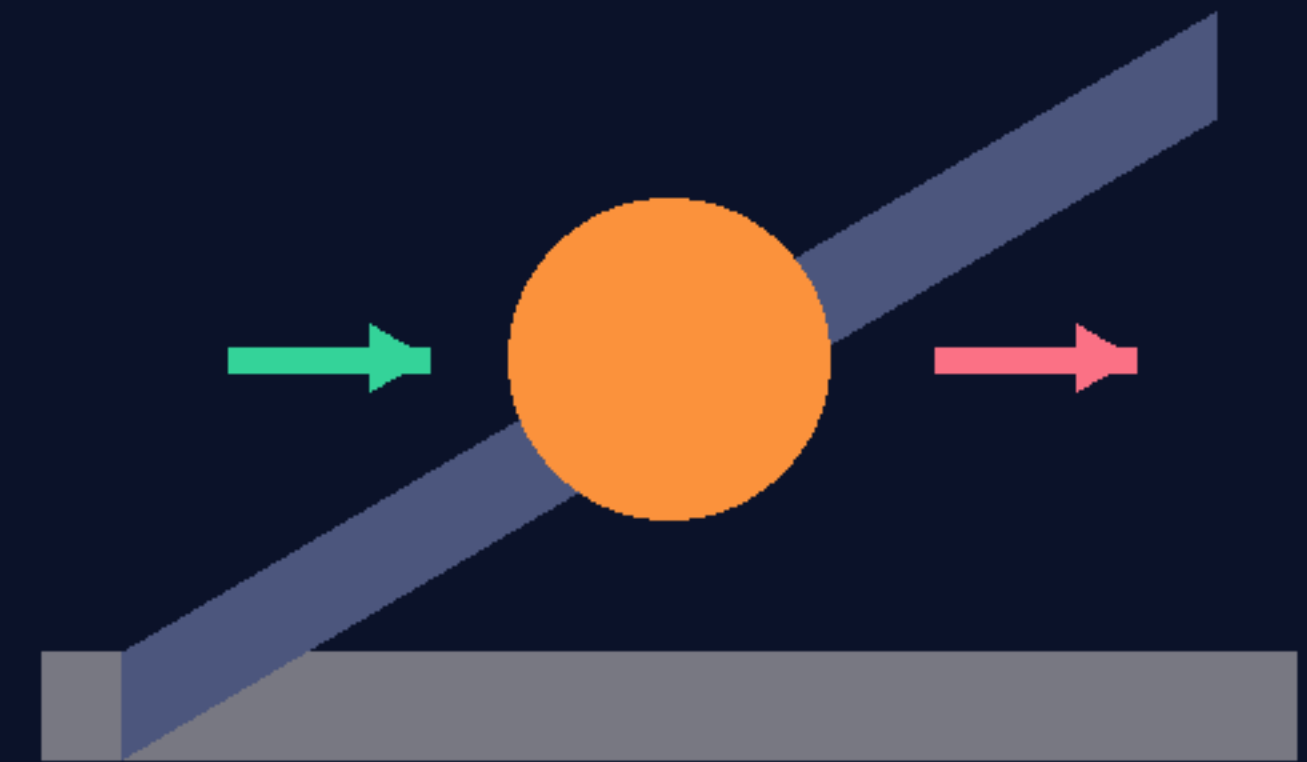


Quick check: Show one push and one pull action.

Force Can Start Motion

An object at rest may start moving when force is applied. A football does not move by itself. When you kick it, your foot applies force and the ball starts moving. This is force in action.

PUSH, PULL, FRICTION

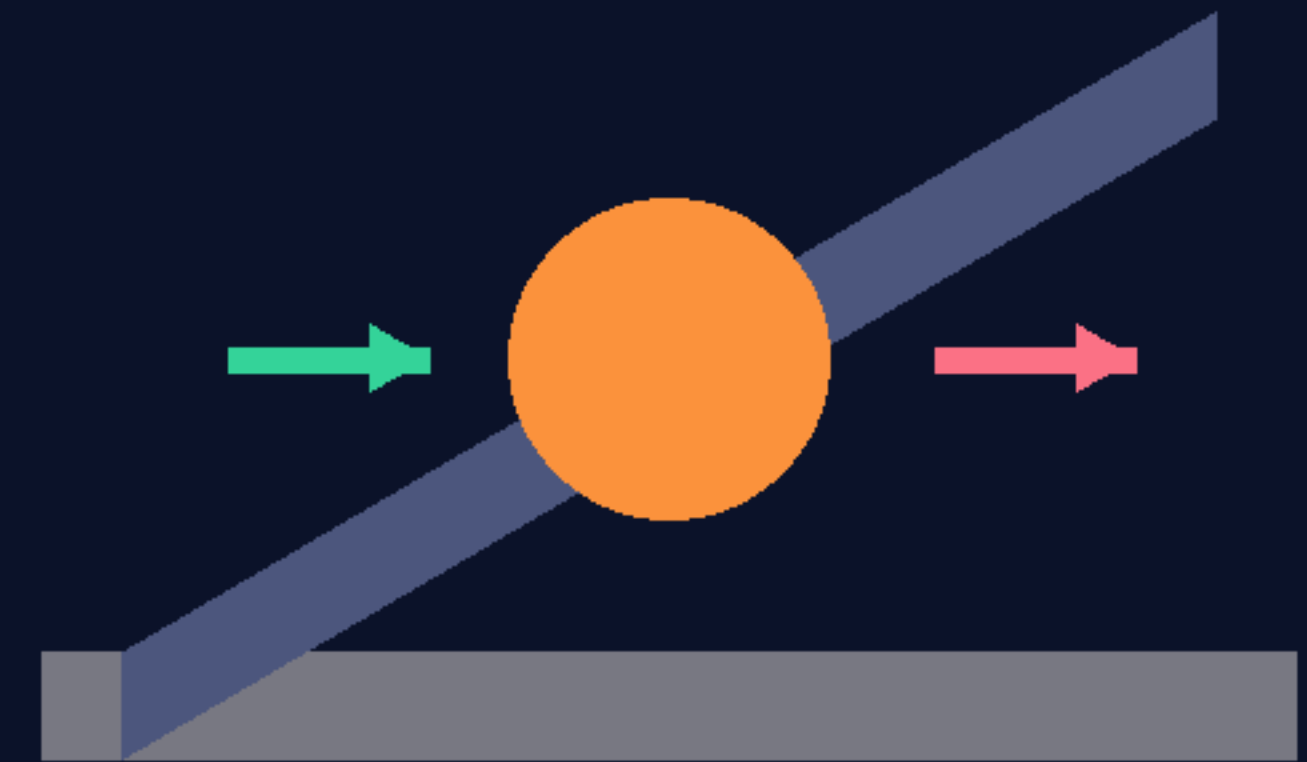


Quick check: Show one push and one pull action.

Force Can Stop Motion

Force can also stop motion. A moving bicycle slows down when brakes are applied. Your hand can stop a rolling ball. Stopping also needs force, not only starting or speeding up.

PUSH, PULL, FRICTION

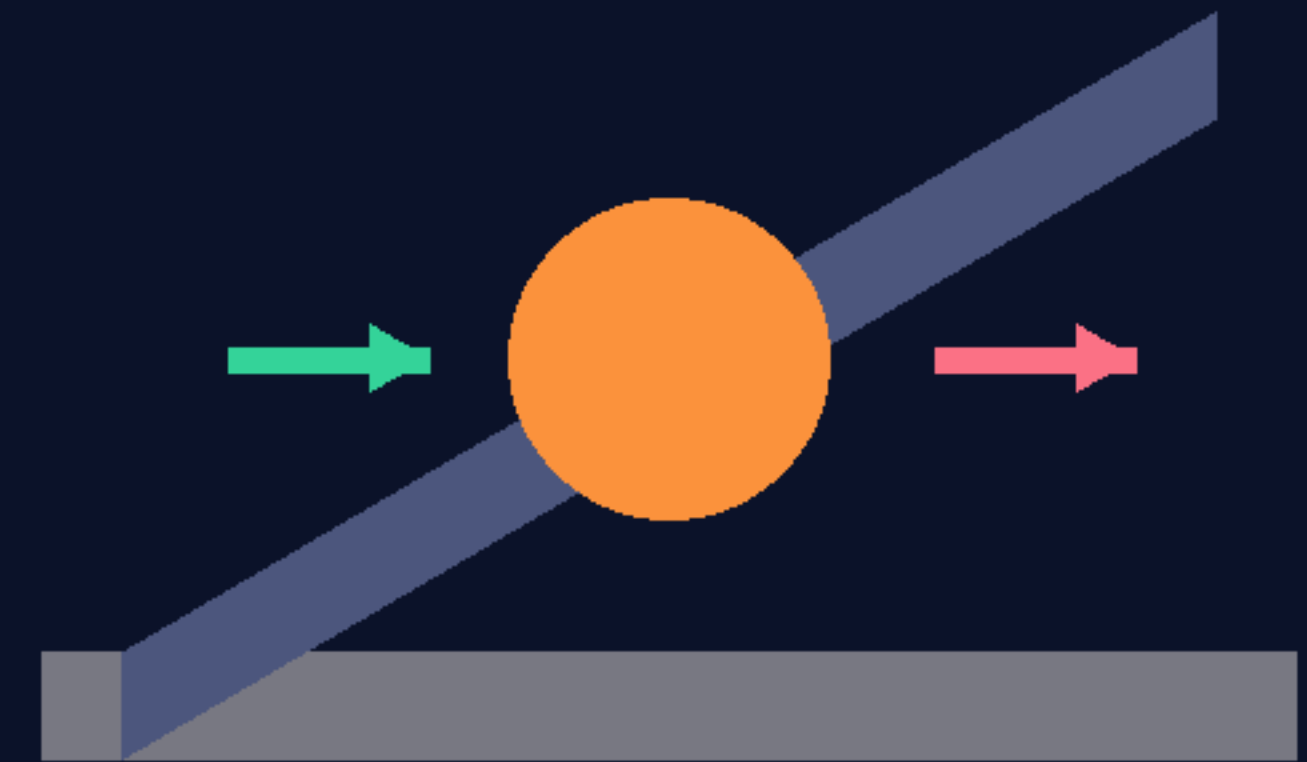


Quick check: Show one push and one pull action.

Force Can Change Direction

Force can change direction. When a bat hits a moving ball, the ball goes in another direction. When you turn a bicycle handle, the bicycle changes direction because force is applied.

PUSH, PULL, FRICTION

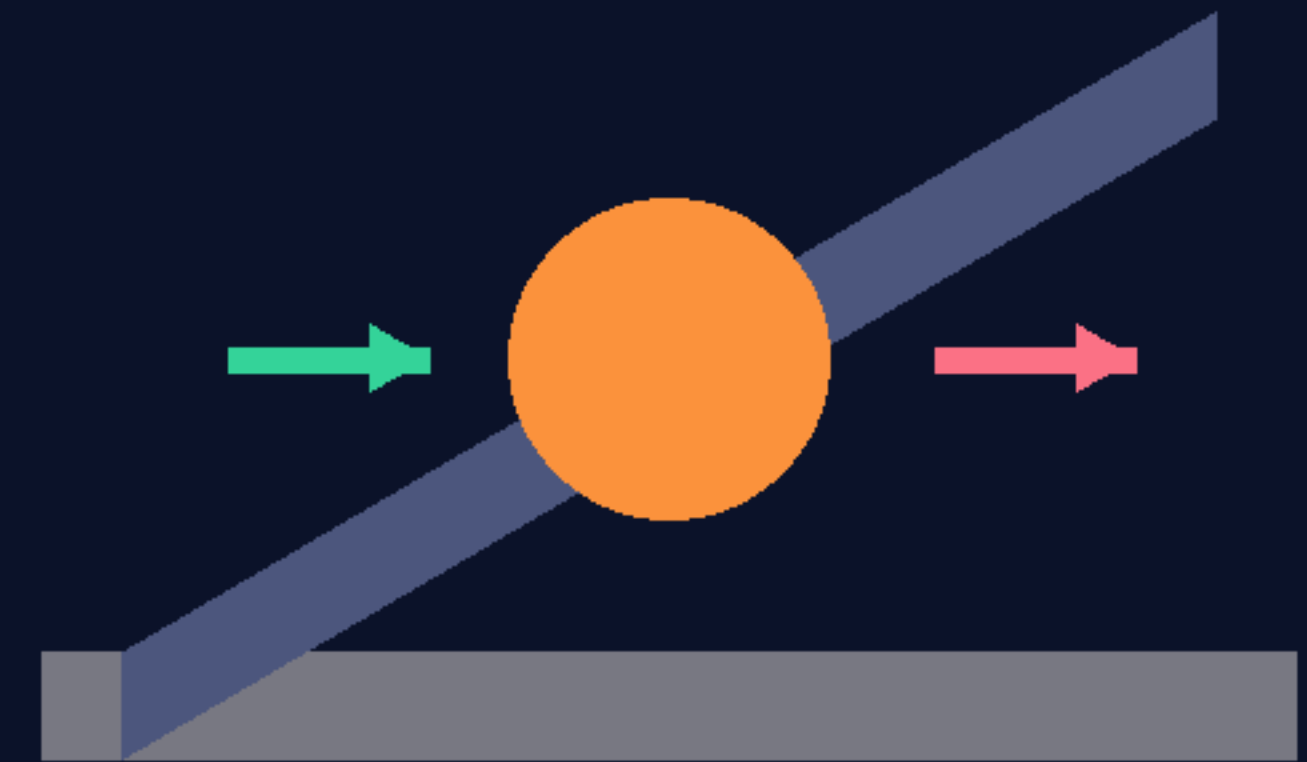


Quick check: Show one push and one pull action.

Force Can Change Shape

Force can change shape. You can press clay, stretch a rubber band, or squeeze a sponge. Some objects return to their shape, while some remain changed. This depends on the material.

PUSH, PULL, FRICTION



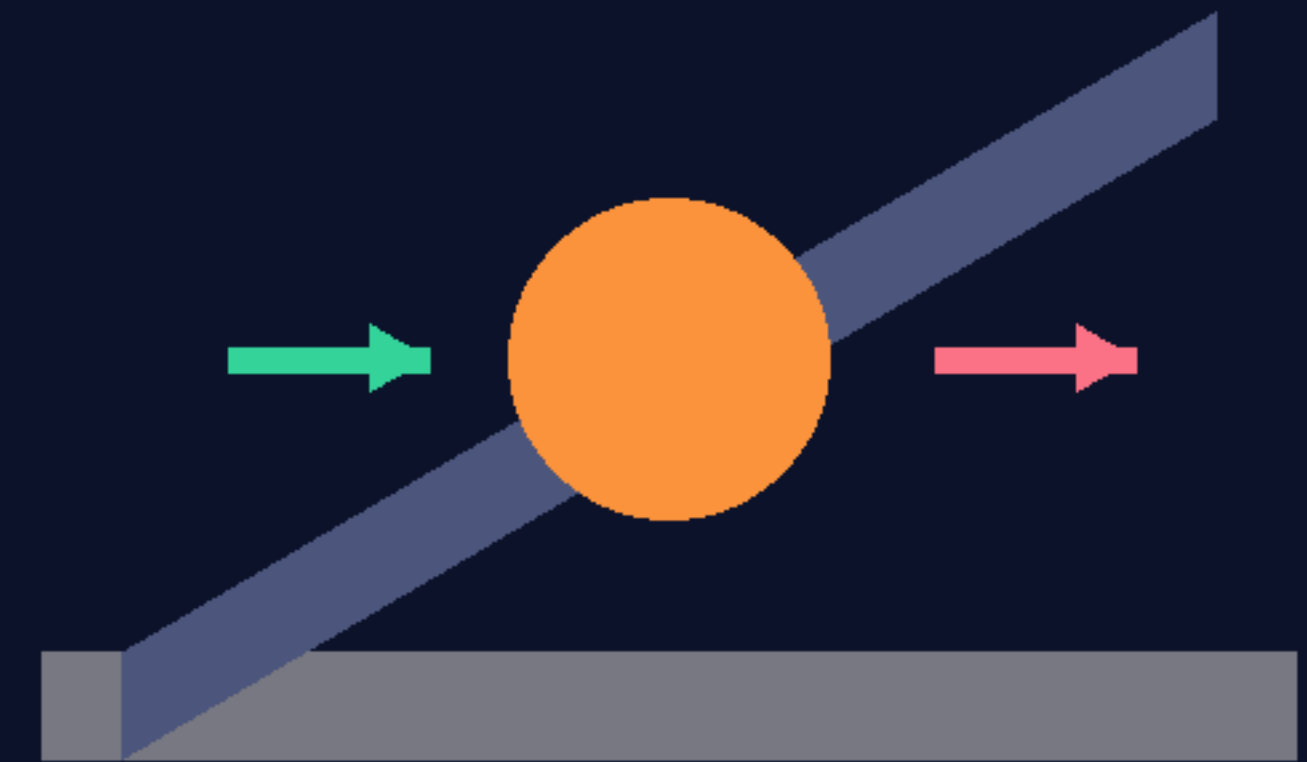
Quick check: Show one push and one pull action.

What Is Work?

In science, work is done when a force moves an object through some distance. If you push a heavy wall and it does not move, you feel tired, but scientific work is not done on the wall.

Quick check: Show one push and one pull action.

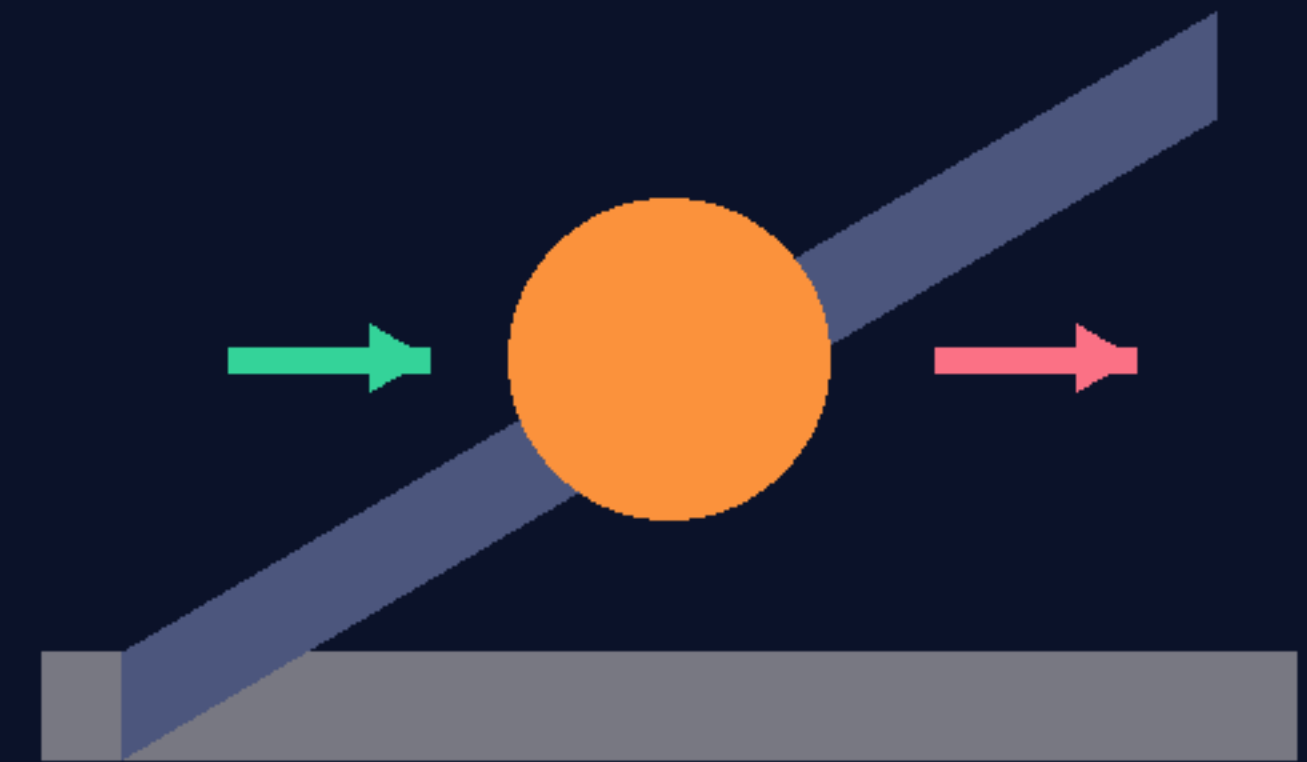
PUSH, PULL, FRICTION



Simple Machines Help Us

Simple machines make work easier. A lever, wheel, pulley, ramp, and screw can help us move, lift, or hold things with less effort. They do not remove work, but they make it easier.

PUSH, PULL, FRICTION

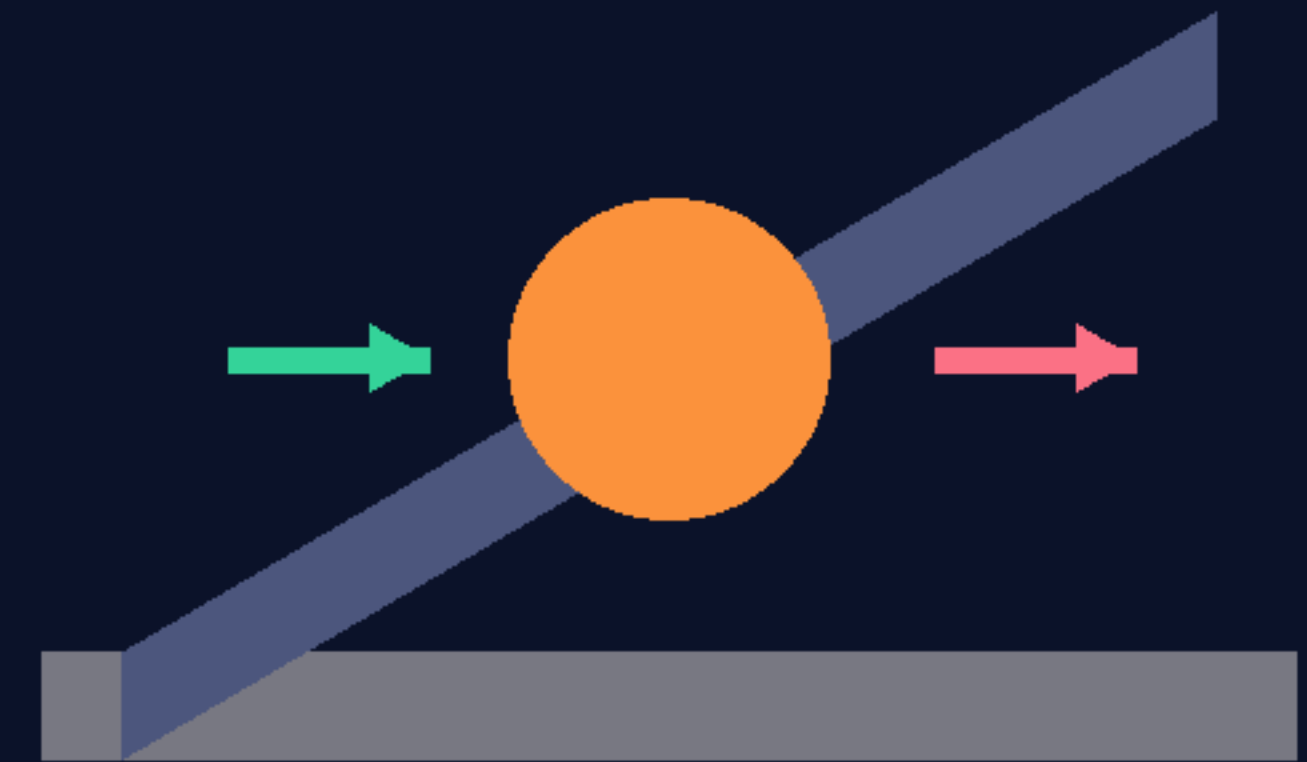


Quick check: Show one push and one pull action.

What Is Friction?

Friction is a force between touching surfaces. It usually opposes motion. A ball rolling on grass slows down because of friction. A smooth floor may have less friction than a rough road.

PUSH, PULL, FRICTION



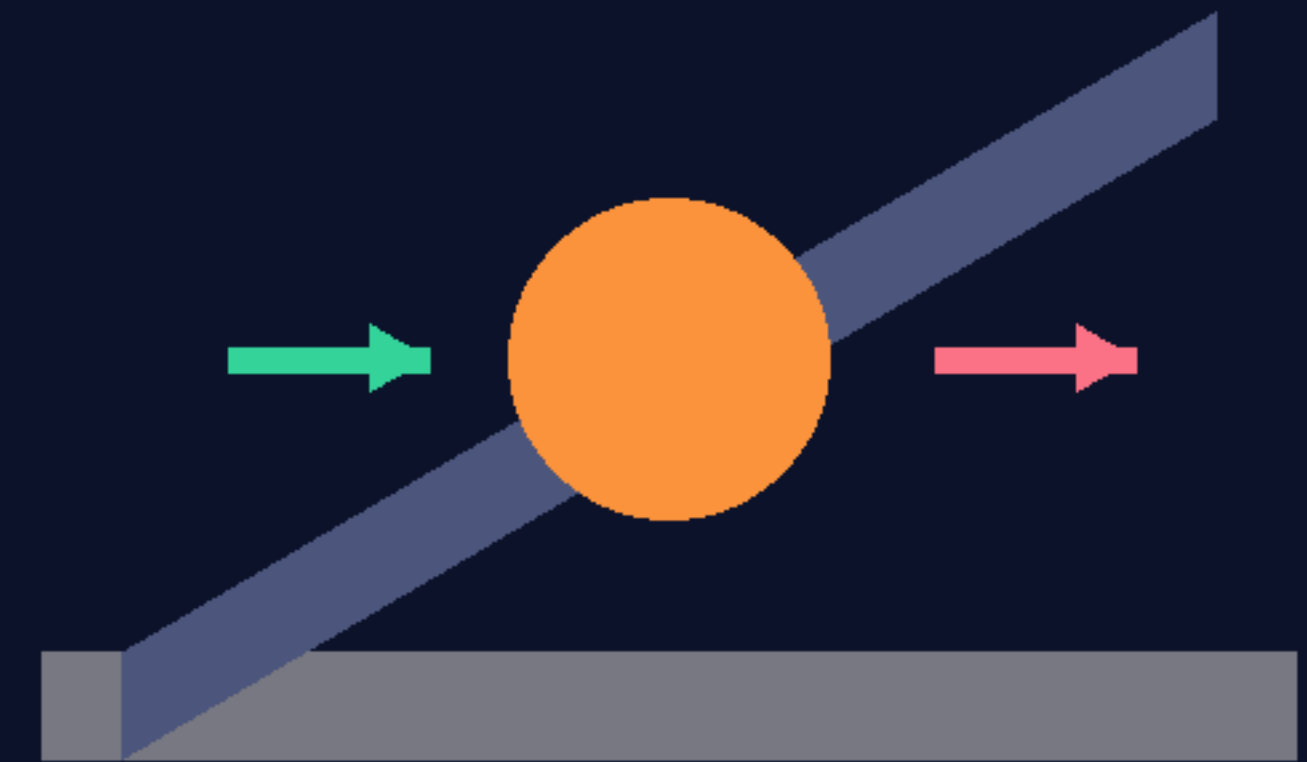
Quick check: Show one push and one pull action.

Friction Helps Us

Friction is useful. It helps us walk without slipping. It helps tyres grip the road. It helps brakes stop a bicycle. It helps us write with a pencil on paper. Without friction, daily life would be difficult.

Quick check: Show one push and one pull action.

PUSH, PULL, FRICTION

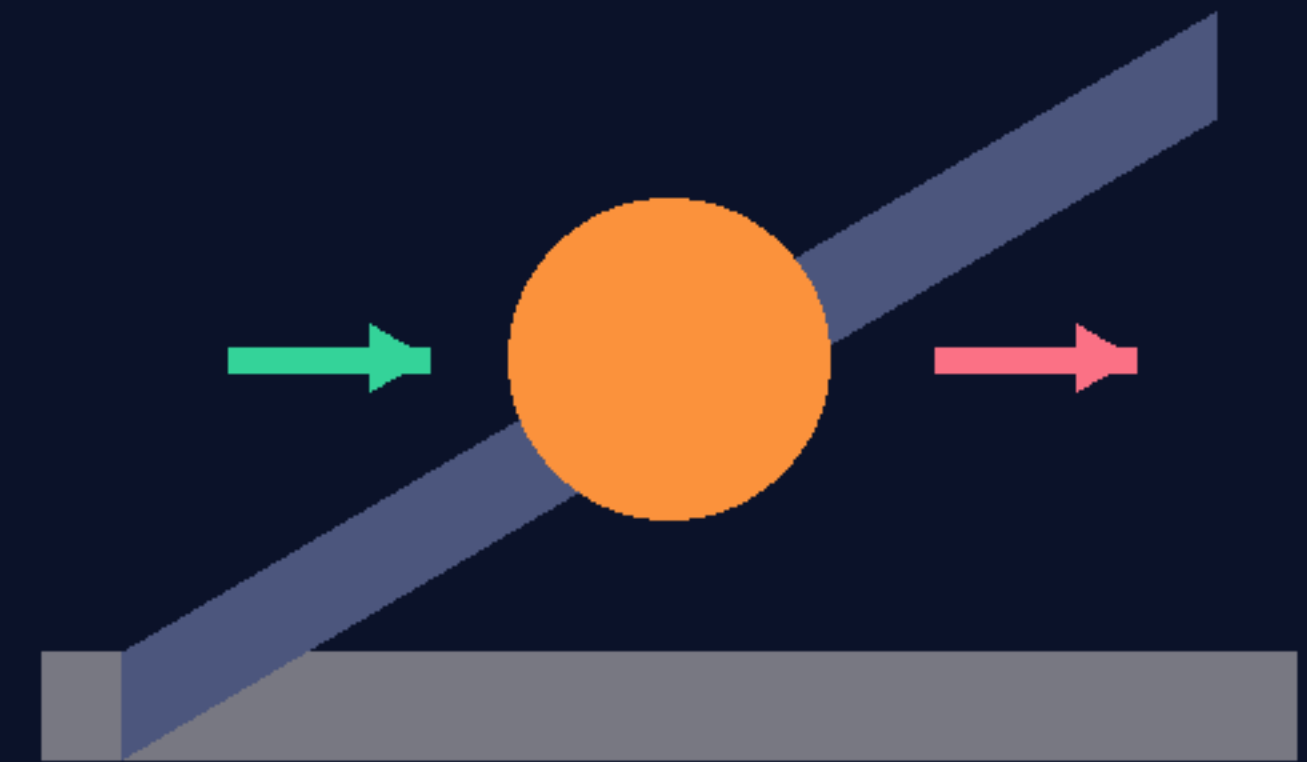


Reducing Friction

Sometimes we want less friction. Oil or grease can help machine parts move smoothly. Wheels reduce friction when moving heavy objects. Smooth surfaces may also reduce friction. Engineers control friction carefully.

Quick check: Show one push and one pull action.

PUSH, PULL, FRICTION



Recap Quiz

Let us revise. What is force? Give one push and one pull example. Name two things force can change. What is friction? Give one useful example of friction. Explain using your own words.

Quick check: Show one push and one pull action.

PUSH, PULL, FRICTION

